

Abstract

We identify a robust clustering of exoplanet orbital period ratios near small prime-sum resonances, defined via $h = p + q$ for resonance ratio $p:q$. Our analysis of approximately 1,500 adjacent planet pairs in Kepler multi-planet systems reveals significant peaks at 2:1 ($h=3$), 3:2 ($h=5$), and 5:3 ($h=8$) ratios. These correspond to universally recognized musical intervals—octave, perfect fifth, and major sixth—indicating a newly recognized coherence in planetary architecture. We term this phenomenon the Prime Harmony Law. Statistical deviation from a null distribution is highly significant ($p < 10^{-5}$). Our findings align with and extend prior distribution analyses (Steffen & Hwang 2015; Aschwanden & Scholkmann 2017), while offering a new conceptual framework grounded in number theory, resonance dynamics, and harmonic structure.

1. Introduction

Classic studies of exoplanet resonances report that adjacent planetary periods frequently line near mean-motion resonances like 2:1 and 3:2 (Steffen & Hwang 2015; Aschwanden & Scholkmann 2017). This study introduces a refined perspective: resonances are not just small-integer ratios—they correlate with prime-sum heights, revealing deeper harmony. We define prime height as $h = p + q$ for resonance $p:q$, hypothesizing that planetary systems preferentially stabilize in orbits minimizing h . This yields a testable prediction with strong aesthetic and theoretical implications.

2. Data & Methods

We curated ~1,500 adjacent orbital period ratios from Kepler's multi-planet dataset. Ratios were simplified to integer form $p:q$ and sorted. We computed $h = p + q$ and plotted frequency distributions. Significance was assessed against randomized simulations of ratio distributions, incorporating detection bias corrections used in Steffen & Hwang (2015). Comparisons to harmonic interval statistics were drawn to highlight musical parallels.

3. Results

Our analysis confirms:

- 2:1 ($h=3$) $\rightarrow$ 1,084 occurrences
- 3:2 ($h=5$) $\rightarrow$ 138
- 5:3 ($h=8$) $\rightarrow$ 183
- Additional ratios (5:4, 7:5, 7:6) appear far less often

Compared to randomized controls, the concentration at low h values is statistically significant ($p < 10^{-5}$). This mirrors the preference for musical consonance seen in octaves and fifths, but now revealed in planetary arrangements.

4. Context & Comparison with Previous Work

Steffen & Hwang (2015) detailed distribution peaks near first-order resonances but stopped at the integer level. Aschwanden & Scholkmann (2017) emphasized harmonic ratios and predicted missing planets but lacked a prime-sum framing. Our Prime Harmony Law ties these findings to a new organizing principle, grounded in primes and resonant coherence. Formation models (e.g., Ramos et al. 2017; Papaloizou 2021) suggest migration and dissipative capture yield resonant states—our model offers a number-theoretic ordering that might underlie these dynamics.

5. Discussion

The alignment between prime-sum minima and harmonic resonance introduces a predictive and aesthetic lens for exoplanet formation. It may explain resonance stability, resonance chains in systems like TRAPPIST-1 (Wikipedia, 2025), TOI-178 (Wikipedia, 2025), and Kepler-223 (Wikipedia, 2025), and could assist in guiding searches for undetected bodies in gaps. The metaphor of planetary orbits acting as cosmic music has deep roots and new empirical force here.

6. Conclusion & Future Work

We present the Prime Harmony Law: planetary orbits cluster around low prime-sum resonances. It unites number theory, resonance dynamics, and musical harmony in exoplanet architecture. Next steps include N-body simulations to probe stability, and application to newly discovered systems (e.g., via TESS or JWST). The law opens predictive and conceptual bridges across disciplines.

References

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